

Based on an extensive analysis of previous year question (PYQ) trends, the latest CBSE blueprint, and the official 2025-26 sample papers [1-3], here is a highly probable, most-expected prediction for the **CBSE Class 12 Physics 2027 Board Exam (70 Marks, 33 Questions)**.

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. Two point charges placed in a medium of dielectric constant K at a distance r experience a force F . What will be the electrostatic force between them in a vacuum at the same distance? (Chapter: Electric Charges and Fields) | **[Confidence: Very High]** [4]

Q2. Which of the following transitions in a hydrogen atom corresponds to the emission of radiation with the maximum wavelength? (a) $n = 4$ to $n = 2$ (b) $n = 3$ to $n = 2$ (c) $n = 2$ to $n = 1$ (d) $n = 4$ to $n = 3$. (Chapter: Atoms) | **[Confidence: Very High]** [5]

Q3. An electron is released from rest in a region of uniform electric and magnetic fields acting parallel to each other. The trajectory of the electron will be: (Chapter: Moving Charges and Magnetism) | **[Confidence: High]** [6]

Q4. The ratio of the nuclear densities of two nuclei having mass numbers 8 and 27 is: (Chapter: Nuclei) | **[Confidence: Very High]** [7]

Q5. Electromagnetic waves used as a diagnostic tool in medicine (or eye surgery) are: (Chapter: Electromagnetic Waves) | **[Confidence: High]** [8]

Q6. The instantaneous values of emf and current in a series AC circuit are $E = E_0 \sin(\omega t)$ and $I = I_0 \sin(\omega t + \pi/3)$. The circuit is necessarily: (Chapter: Alternating Current) | **[Confidence: Medium]** [9]

Q7. A charged particle after being accelerated through a potential difference V enters a uniform magnetic field and moves in a circle of radius r . If V is doubled, the radius of the circle will become: (Chapter: Moving Charges and Magnetism) | **[Confidence: High]** [6]

Q8. In an extrinsic semiconductor, the formation of the depletion region in a p-n junction diode is strictly due to: (Chapter: Semiconductor Electronics) | **[Confidence: Very High]** [10]

Q9. The shape of the interference fringes in Young's double-slit experiment when D (distance between slit and screen) is very large compared to fringe width is nearly: (Chapter: Wave Optics) | **[Confidence: Medium]** [11]

Q10. The work function for a metal surface is 4.14 eV . The threshold wavelength for this metal surface is: (Chapter: Dual Nature of Radiation & Matter) | **[Confidence: High]** [12]

Q11. If an ammeter is to be used in place of a voltmeter, then we must connect with the ammeter a: (Chapter: Moving Charges and Magnetism) | **[Confidence: High]** [13]

Q12. If the frequency of an AC circuit is made 4 times its initial value, the inductive reactance will be: (Chapter: Alternating Current) | **[Confidence: High]** [9]

Q13. Assertion (A): Work done in moving a charge around a closed path in an electric field is always zero. **Reason (R):** Electrostatic force is a conservative force. (*Chapter: Electrostatics*) | **[Confidence: Very High]** [14]

Q14. Assertion (A): The critical angle of light passing from glass to air is minimum for violet colour. **Reason (R):** The wavelength of blue light is greater than the light of other colours. (*Chapter: Ray Optics*) | **[Confidence: High]** [15]

Q15. Assertion (A): The electrical conductivity of a semiconductor increases on doping. **Reason (R):** Doping always increases the number of electrons in the semiconductor. (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]** [10]

Q16. Assertion (A): Two light sources emitting waves of similar wavelengths are coherent. **Reason (R):** Two light sources emitting waves having zero or constant phase difference are known as coherent sources. (*Chapter: Wave Optics*) | **[Confidence: High]** [16]

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. Draw equipotential surfaces due to a single isolated point charge ($-q$) and depict the electric field lines. (*Chapter: Electrostatic Potential and Capacitance*) | **[Confidence: Very High]** [17]

Q18. Name the parts of the electromagnetic spectrum which are (a) suitable for radar systems used in aircraft navigation, and (b) used as a diagnostic tool in medicine. Write one method of production for each. (*Chapter: Electromagnetic Waves*) | **[Confidence: Very High]** [18]

Q19. An alpha particle is accelerated through a potential difference of 100 V . Calculate the de-Broglie wavelength associated with it. (Take mass of alpha particle = $6.4 \times 10^{-27}\text{ kg}$). (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: High]** [19]

Q20. A cell of emf $\mathcal{E}$ and internal resistance r is connected across a variable resistor R . Plot a graph showing the variation of terminal voltage V of the cell versus the current I . Using the plot, explain how the internal resistance can be determined. (*Chapter: Current Electricity*) | **[Confidence: High]** [20]

Q21. Calculate the shortest wavelength in the Balmer series of the hydrogen atom. In which region (infrared, visible, ultraviolet) does this wavelength lie? (*Chapter: Atoms*) | **[Confidence: Medium]** [21]

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. Using Gauss's theorem, deduce an expression for the electric field intensity at a point due to a uniformly charged infinite plane sheet. (*Chapter: Electric Charges and Fields*) |

[Confidence: Very High] [22]

Q23. State Biot-Savart law. Use it to derive the expression for the magnetic field on the axis of a current-carrying circular loop. (*Chapter: Moving Charges and Magnetism*) | **[Confidence: Very High]** [23]

Q24. Draw a labelled diagram of an AC generator. Obtain the expression for the instantaneous value of the induced emf in terms of the magnetic field B , number of turns N of the coil, area A , and angular frequency ω . (*Chapter: Electromagnetic Induction*) | **[Confidence: High]** [24]

Q25. A $2\mu\text{F}$ capacitor, 100Ω resistor, and 8 H inductor are connected in series with an AC source. What should be the frequency of the source such that the current drawn in the circuit is maximum? What is this condition called? (*Chapter: Alternating Current*) | **[Confidence: High]** [25]

Q26. Using Huygens' principle, draw a diagram showing the propagation of a plane wavefront reflecting at a plane surface. Hence, verify the laws of reflection. (*Chapter: Wave Optics*) | **[Confidence: Very High]** [26]

Q27. State Einstein's photoelectric equation. Explain with the help of this equation any two observed features of the photoelectric effect which cannot be explained by classical wave theory. (*Chapter: Dual Nature of Radiation & Matter*) | **[Confidence: Very High]** [27]

Q28. Explain with the help of a labelled circuit diagram the working of a p-n junction diode as a full-wave rectifier. Draw the input and output waveforms. (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]** [28, 29]

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts (1+1+2 marks).

Q29. Case Study 1: Total Internal Reflection / Optical Fibers *Passage:* Propagation of light through an optical fibre is due to total internal reflection taking place at the core-cladding interface. The refractive index of the core is strictly greater than the cladding. [18] (a) Define critical angle. (1 mark) (b) State the two essential conditions for total internal reflection to occur. (1 mark) (c) A ray of light incident at an angle θ on an optical fiber core-cladding boundary just undergoes TIR. Calculate the refractive index of the core if the cladding is air. **OR** Explain how the phenomenon of mirage occurs in deserts. (2 marks) (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: High]** [11]

Q30. Case Study 2: Wheatstone Bridge and Kirchhoff's Rules *Passage:* The Wheatstone bridge is an application of Kirchhoff's rules used to find unknown resistances. It consists of four resistors arranged in a quadrilateral. [30] (a) State Kirchhoff's Junction Rule and mention the conservation law it is based on. (1 mark) (b) What is the condition for a Wheatstone bridge to be "balanced"? (1 mark) (c) Derive the balanced condition of a Wheatstone bridge using Kirchhoff's loop rules. **OR** If the galvanometer and battery are interchanged in a

balanced Wheatstone bridge, how will the balance condition be affected? Justify. (2 marks)
(Chapter: Current Electricity) | **[Confidence: Very High]** [29, 30]

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) Derive an expression for the capacitance of a parallel plate capacitor when a dielectric slab of thickness ' t ' ($t < d$) is introduced between its plates, where ' d ' is the separation between the plates. [31] **(b)** Two identical capacitors of **12 pF** each are connected in series across a battery of **50 V**. How much electrostatic energy is stored in the combination? [32] **OR (a)** Derive an expression for the torque acting on an electric dipole placed in a uniform electric field. Identify the orientations for stable and unstable equilibrium. [33] **(b)** Calculate the work done in rotating an electric dipole from its stable equilibrium position to unstable equilibrium. *(Chapter: Electrostatics)* | **[Confidence: Very High]**

Q32. (a) Derive an expression for the force per unit length between two infinitely long straight parallel current-carrying wires. Hence, define exactly "One Ampere" of current. [34] **(b)** Two parallel very long straight wires carrying a current of **5 A** each are kept at a separation of **1 m** in the same direction. What will be the nature and magnitude of the force per unit length between them? [34] **OR (a)** With the help of a labelled diagram, explain the principle, construction, and working of a moving coil galvanometer. [35] **(b)** Explain how a moving coil galvanometer can be converted into an ammeter of a given range. *(Chapter: Moving Charges and Magnetism)* | **[Confidence: Very High]**

Q33. (a) Derive the Lens Maker's Formula $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ for a convex lens. State the sign conventions used. [36] **(b)** A convex lens of focal length **20 cm** is placed coaxially in contact with a concave lens of focal length **25 cm**. Determine the power of the combination. **OR (a)** Derive the expression for the fringe width in Young's Double Slit Experiment (YDSE). [11] **(b)** How will the interference pattern in YDSE be affected if the entire apparatus is immersed in water? Justify. [11] *(Chapter: Optics)* | **[Confidence: Very High]**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.